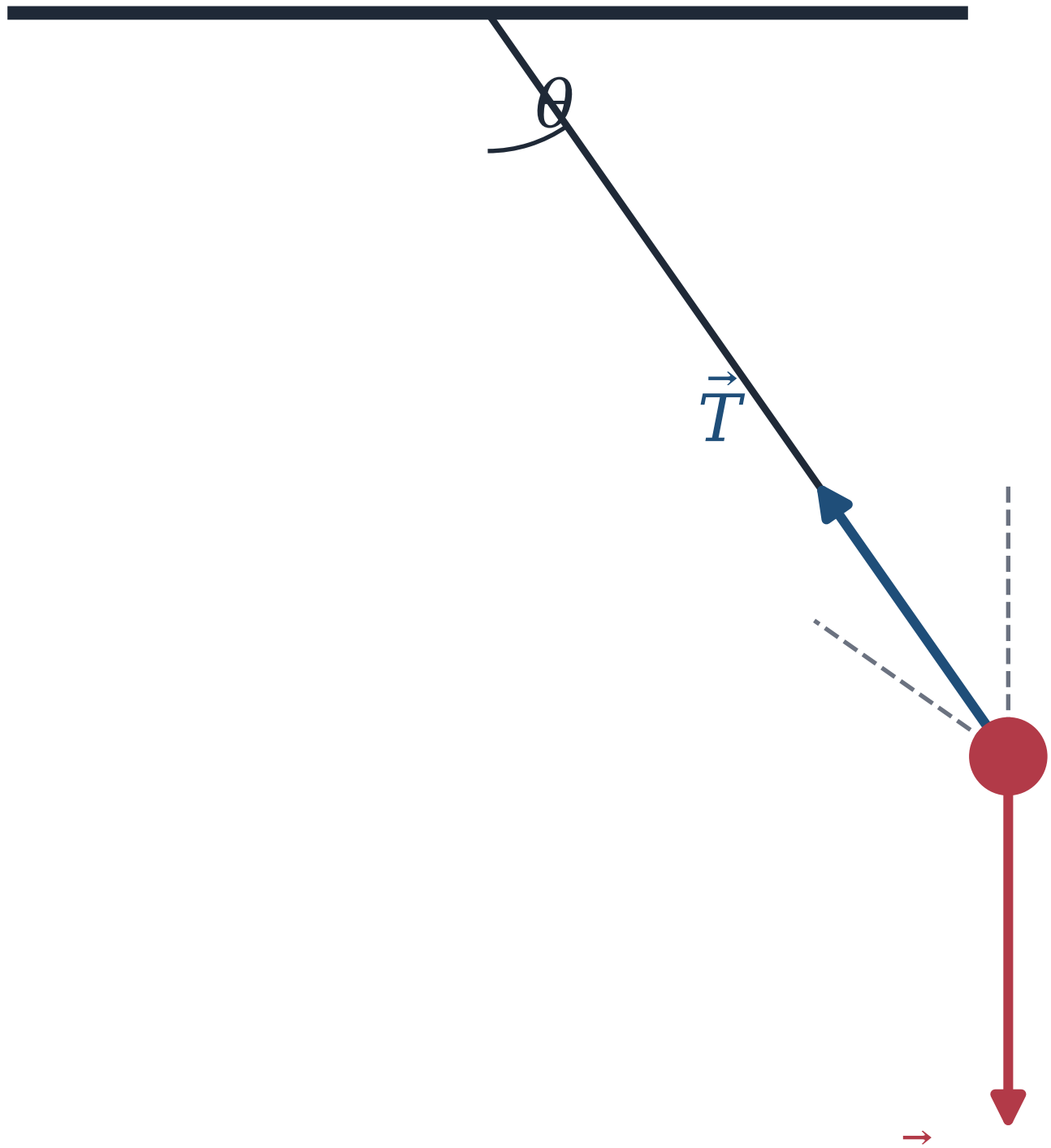


$$\ddot{\theta} + \frac{g}{L}\sin\theta = 0$$